13 March 2026

Exercise 1 : SQL Fundamentals

Database: employee_db

Assume you have a table called "employees" with the following structure :

| id | first_name | last_name | department | salary | hire-date | city |
|---|---|---|---|---|---|---|
| 1 | John | Doe | IT | 55 000 | 2018-06-15 | New York |
| 2 | Jane | Smith | HR | 48 000 | 2019-07-20 | Chicago |
| 3 | Mike | Johnson | Finance | 60 000 | 2017-09-30 | Los Angeles |
| 4 | Sarah | Brown | IT | 53 000 | 2021-03-25 | New York |
| 5 | David | White | Marketing | 52 000 | 2016-04-10 | San Francisco |
| 6 | Emily | Davis | IT | 62 000 | 2015-02-14 | Chicago |
| 7 | Robert | Wilson | Finance | 59 000 | 2019-10-01 | Houston |
| 8 | Jessica | Moore | HR | 51 000 | 2018-05-22 | Los Angeles |
| 9 | Daniel | Clark | Marketing | 53 000 | 2022-06-01 | Chicago |
| 10 | Laura | Hall | IT | 50 000 | 2020-08-10 | San Francisco |

1. SELECT Statement

Write a SQL query to retrieve all columns from the employees table.

Answer :

```
SELECT *
FROM employees;
```

Output: The table above

2. SELECT DISTINCT Statement

Write a SQL query to find all the unique departments in the employees table.

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Answer : SELECT DISTINCT department
FROM employees;

Output

| department |
|---|
| IT |
| HR |
| Finance |
| Marketing |

3. ORDER BY Statement

Write a SQL query to retrieve all employees first and last names, ordered by salary in descending order.

Answer : SELECT first_name,
last_name,
salary
FROM employees
ORDER BY salary DESC;

Output :

| first_name | last_name | Salary |
|---|---|---|
| Emily | Davis | 62000 |
| Mike | Johnson | 60000 |
| Robert | Wilson | 59000 |
| John | Doe | 55000 |
| Sarah | Brown | 53000 |
| Daniel | Clark | 53000 |
| David | White | 52000 |
| Jessica | Moore | 51000 |
| Laura | Hall | 50000 |
| Jane | Smith | 48000 |

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4. LIMIT Statement

Write a SQL query to retrieve the top 5 highest-paid employees.

Answer :
```
SELECT first_name,
       last_name,
       Salary
FROM employees
ORDER BY salary DESC
LIMIT 5;
```

Output:

| first_name | last_name | Salary |
|---|---|---|
| Emily | Davis | 62000 |
| Mike | Johnson | 60000 |
| Robert | Wilson | 59000 |
| John | Doe | 55000 |
| Sara | Brown | 53000 |

5. WHERE Statement

Write a SQL query to find employees who work in the IT department.

Answer :
```
SELECT first_name,
       last_name,
       department
FROM employees
WHERE department = 'IT';
```

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6. AND Statement

Write a SQL query to find employees who work in the Finance department AND have a salary greater than 58,000

Answer:
```
SELECT first_name,
       last_name,
       department,
       Salary
FROM employees
WHERE department = 'Finance' AND salary > 58000;
```

Output:

| first_name | last_name | department | Salary |
|---|---|---|---|
| Mike | Johnson | Finance | 60000 |
| Robert | Wilson | Finance | 59000 |

7. OR Statement

Write a SQL query to find employees who work in the HR department OR the Marketing department.

Answer:
```
SELECT first_name,
       last_name,
       department
FROM employees
WHERE department = 'HR' OR department = 'Marketing';
```

Output:

| first_name | last_name | department |
|---|---|---|
| Jane | Smith | HR |
| David | White | Marketing |
| Jessica | Moore | HR |
| Daniel | Clark | Marketing |

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8. NOT Statement

Write a SQL query to find employees who do not work in the IT department

Answer:
```
SELECT first_name,
       last_name,
       department
FROM employees
WHERE NOT department = 'IT';
```

Output:

| first_name | last_name | department |
|---|---|---|
| Jane | Smith | HR |
| Mike | Johnson | Finance |
| David | White | Marketing |
| Robert | Wilson | Finance |
| Jessica | Moore | HR |
| Daniel | Clark | Marketing |

9. IN Statement

Write a SQL query to find employees who are in HR, IT, or Finance departments.

Answer:
```
SELECT first_name,
       last_name,
       department
FROM employees
WHERE department IN('HR', 'IT', 'Finance')
```

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Output:

| first_name | last_name | department |
|---|---|---|
| John | Doe | IT |
| Jane | Smith | HR |
| Mike | Johnson | Finance |
| Sarah | Brown | IT |
| Emily | Davis | IT |
| Robert | Wilson | Finance |
| Jessica | Moore | HR |
| Laura | Hall | IT |

10. Combining conditions

Write a SQL query to find employees who are in the IT department, have a salary greater than 50,000 and are located in New York.

Answer:
```
SELECT first_name,
       last_name,
       department,
       salary,
       city
FROM employees
WHERE city IN('New York') AND department = 'IT' AND salary
       > 50000;
```

Output:

| first_name | last_name | department | salary | city |
|---|---|---|---|---|
| John | Doe | IT | 55000 | New York |
| Sarah | Brown | IT | 53000 | New York |

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11. Combining WHERE, AND, and ORDER BY

Write a SQL query to retrieve the first and last names of employees who work in the Finance, or Marketing departments, earn more than 52000, and order the results by salary in descending order

Answer:
```
SELECT first_name,
       last_name,
       department,
       salary
FROM employees
WHERE department = 'Finance' OR department = 'Marketing'
AND salary > 52000
ORDER BY salary DESC;
```

Output:

| first_name | last_name | department | Salary |
|---|---|---|---|
| Mike | Johnson | Finance | 60000 |
| Robert | Wilson | Finance | 59000 |
| Daniel | Clark | Marketing | 53000 |

12. Combining SELECT DISTINCT, WHERE and IN

Write a SQL query to find all the unique cities where employees work, excluding those in the IT and HR departments.

Answer:
```
SELECT DISTINCT city,
                department
FROM employees
WHERE department IN ('Finance', 'Marketing');
```

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Output :

| City | department |
|---|---|
| Los Angeles | Finance |
| San Francisco | Marketing |
| Houston | Finance |
| Chicago | Marketing |

13. Combining WHERE, NOT, AND, and ORDER BY

Write a SQL query to retrieve employees who are NOT in the Finance department, have a salary greater than 50 000, and order the results by hire date in ascending order.

Answer:
```
SELECT first_name,
       last_name,
       department,
       salary,
       hire_date
FROM employees
WHERE NOT department = 'Finance' AND salary > 50000
ORDER BY hire_date ASC;
```

Output:

| first_name | last_name | department | salary | hire_date |
|---|---|---|---|---|
| Emily | Davis | IT | 62000 | 2015-02-14 |
| David | White | Marketing | 52000 | 2016-04-10 |
| Jessica | Moore | HR | 51000 | 2018-05-22 |
| John | Doe | IT | 55000 | 2018-06-15 |
| Sarah | Brown | IT | 58000 | 2021-03-25 |
| Daniel | Clark | Marketing | 53000 | 2022-06-01 |

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14. Combining WHERE, OR, IN, and LIMIT

Write a SQL query to find the first 3 employees who work in either Chicago or Los Angeles and belong to the IT or Marketing department.

Answer:
```
SELECT first_name,
       last_name,
       city,
       department
FROM employee
WHERE city IN ('Chicago', 'Los Angeles') AND
department = 'IT' OR department = 'Marketing'
LIMIT 3;
```

Output:

| first_name | last_name | city | department |
|---|---|---|---|
| David | White | San Francisco | Marketing |
| Emily | Davis | Chicago | IT |
| Daniel | Clark | Chicago | Marketing |

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